**Program - 1** - Design and implementation of a HALF GDDER using VHDL

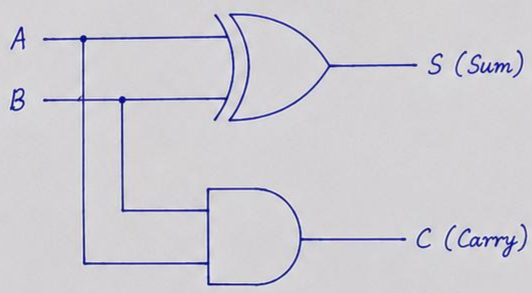
⇒ Inputs : A, B

⇒ Output : S (Sum), C (Carry)

⇒ Truth table :

| A | B | S | C |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

Circuit :

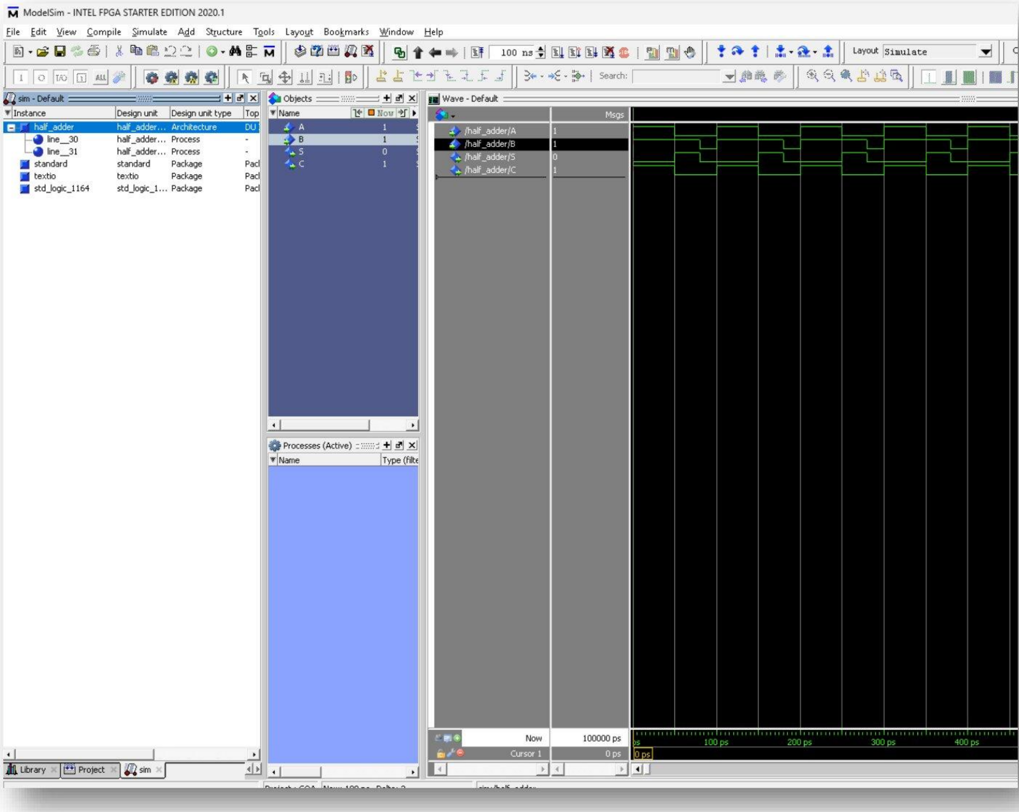



⇒ Program :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity half_adder is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        S : out STD_LOGIC;    -- Sum output
        C : out STD_LOGIC     -- Carry output
    );
end half_adder;

architecture Behavioral of half_adder is
begin
    S <= A XOR B;    -- Sum = A XOR B
    C <= A AND B;    -- Carry = A AND B
end Behavioral;
```



***Simulation Output of Half Adder***

# Assignment - 1

## Program - 1 - Design and implementation of a HALF SUBTRACTOR using VHDL

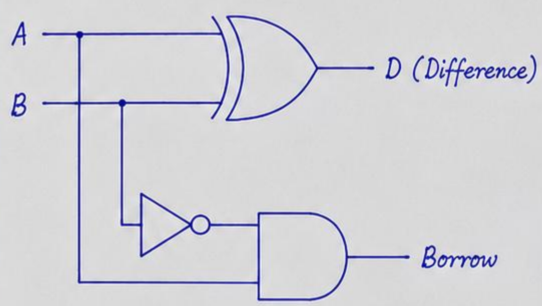
⇒ Inputs : A, B

⇒ Outputs : D (Difference), Borrow

⇒ Truth table :

| A | B | D | Borrow |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |

Circuit :

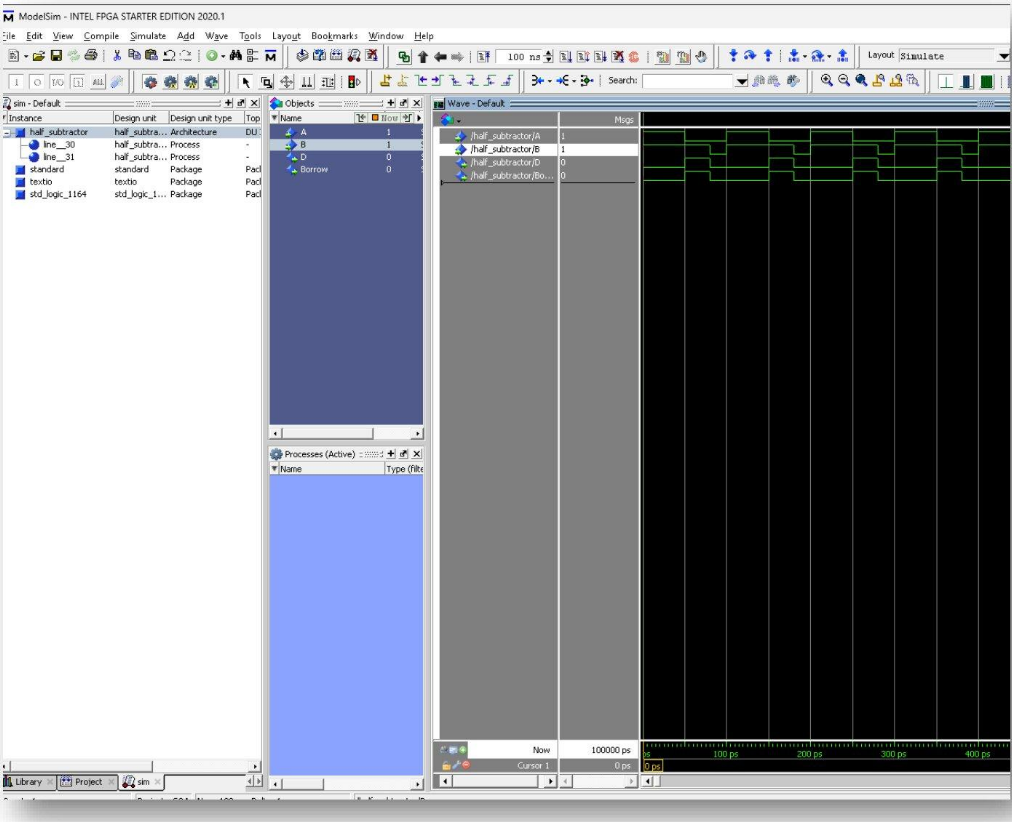



⇒ Program :

```
Library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity half_subtractor is
    Port (
        A      : in  STD_LOGIC;    -- Minuend
        B      : in  STD_LOGIC;    -- Subtrahend
        D      : out STD_LOGIC;    -- Difference output
        Borrow : out STD_LOGIC     -- Borrow output
    );
end half_subtractor;

architecture Behavioral of half_subtractor is
begin
    D      <= A XOR B;              -- Difference = A XOR B
    Borrow <= (NOT A) AND B;        -- Borrow = (NOT A) AND B
end Behavioral;
```




***Simulation Output of Half Subtractor***